Marburg University

Department of Computer Science
Prof. Dr. Sebastian Wild

Date: 2026-10-30
Version: 2026-10-30 16:09

Exercise Sheet 3 for Advanced Algorithms (Winter 2026/27)

Hand In: *Until 2026-11-06 18:00, on ILIAS.*

Problem 1

30 points

In class, we have seen that the *p-variable*-3SAT problem, i.e., the 3CNF satisfiability problem parametrized by the number k of variables trivially satisfies *p-variable*-3SAT $\in \mathcal{FPT}$ since the $\mathcal{O}(2^k n)$ brute-force algorithm is an fpt-algorithm for this problem. But that algorithm actually solves the general SAT problem – indeed, even the satisfiability problem for boolean *circuits*.

Show that we can indeed exploit the special structure of 3SAT to improve upon this by designing an fpt-algorithm for *p-variable*-3SAT with search space $\mathcal{O}(\alpha^k)$ for some $\alpha < 2$.

Hint: Find a *disjoint* decomposition of the search space of all assignments that satisfy the first clause $l_1 \vee l_2 \vee l_3$.